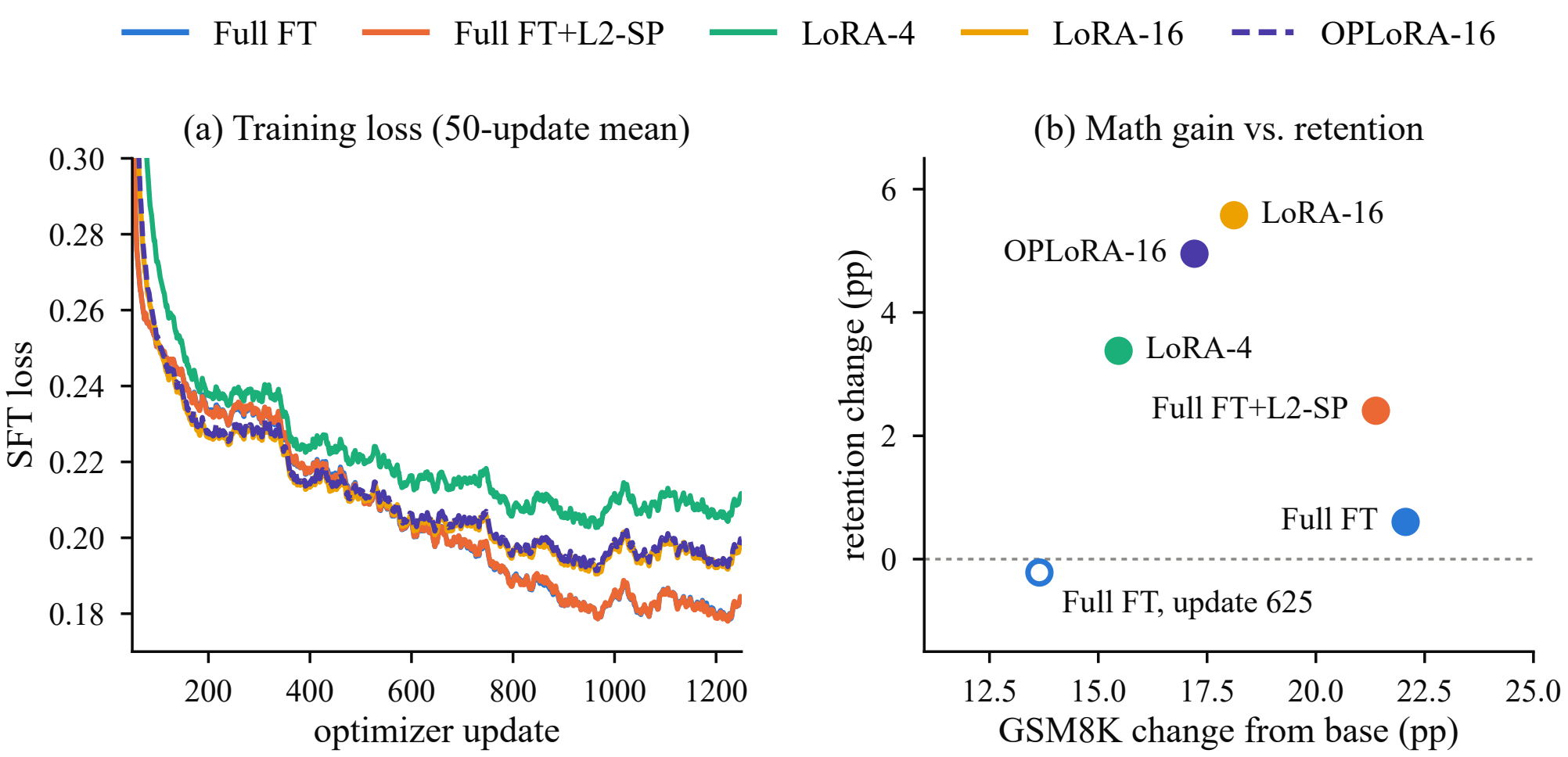

Full FT
Full FT+L2-SP
LoRA-4
LoRA-16
OPLoRA-16
(a) Training loss (50-update mean)
SFT loss
0.30
0.28
0.26
0.24
0.22
0.20
0.18
200
400
600
800
1000
1200
optimizer update
(b) Math gain vs. retention
retention change (pp)
6
4
2
0
LoRA-16
OPLoRA-16
LoRA-4
Full FT+L2-SP
Full FT
Full FT, update 625
12.5
15.0
17.5
20.0
22.5
25.0
GSM8K change from base (pp)